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Analytic Input Convex Neural Networks-based Model Predictive Control for Power System Transient Stability Enhancement

Tong Su, *Student Member, IEEE*, Junbo Zhao, *Senior Member, IEEE*, Xiao Chen, Xiaodong Liu

Abstract—Transient stability-constrained optimal power flow (TSC-OPF) is a non-convex optimization problem and also computationally intensive to solve. To find the global optimum and improve the convergence, this paper proposes an analytic input convex neural networks (ICNN)-based model predictive control (MPC) for transient stability enhancement. ICNN is developed to predict power system transient stability that has the particular property of ensuring its output is convex to the input. This capability allows ICNN to optimize the convex inputs, such as the active power output of generators, while keeping other values constant, such as variables that do not participate in optimization. In addition, non-convex static constraints are built by decoupled linearized power flow model to form a fully convex TSC-OPF with a global optimum. The proposed method is embedded into MPC to enhance transient stability. Numerical results show that the proposed method can greatly improve convergence, find the global optimal solution, and speed up the calculation.

Index Terms—Transient stability constrained optimal power flow, input convex neural networks, model predictive control, power system transient stability, decoupled linearized power flow.

I. INTRODUCTION

TRANSIENT stability constrained optimal power flow (TSC-OPF) is to find the optimal operating point of the power system subject to system stability constraints. Due to the non-linear and non-convex constraints, the TSC-OPF is NP-hard (Non-deterministic Polynomial-time hardness) and cannot guarantee convergence and global optimum [1], [2].

For static stability, the most accurate solution method is AC optimal power flow (AC-OPF), where constraints are non-linear and non-convex. To speed up the calculation, DC optimal power flow (DC-OPF), which linearizes the power flow constraints, is proposed and dramatically simplifies the analysis and guarantees convergence. However, DC-OPF cannot calculate the bus voltage amplitude and reactive power flow [3]. Therefore, improved DC-OPF methods are proposed, such as hot- and cold- start models. Hot-start models use an initial AC power flow solution to correct bus injections. Whenever the initial AC solutions are not available, such as

in DC-OPF-based unit commitment or long-term planning, the cold-start models can be used by setting the net loss as a percentage of the load or fixing the bus voltage magnitudes at either nominal or typical values [4]. Based on cold-start models, a novel DC power flow method is proposed in [5] to consider reactive power balance equations. Additionally, a fully decoupled linearized power flow (DLPF) model is proposed in [3], which achieves the high accuracy of voltage magnitudes and robust convergence in solving meshed and large-scale transmission systems.

For transient stability, the most accurate method is a time-domain simulation by solving the differential-algebraic equations (DAEs) [6]. Many alternative methods are proposed to verify the transient stability, such as quasi-steady-state analysis [7], transient energy function method [8], extended equal-area criterion [9], etc. Unfortunately, those model-based methods are still non-linear and non-convex, which makes it difficult to obtain robust solutions and apply the methods to on-line optimization. In recent years, artificial intelligence-based methods are proposed to improve the calculation efficiency [6], [10], [11]. However, most of the existing artificial intelligence methods are inherently nonlinear and non-convex. All the above methods may result in local optima or convergence failure in optimization [12], [13].

In this paper, an analytic input convex neural networks (ICNN)-based model predictive control (MPC) for enhancing power system transient stability is proposed. The main contributions are as follows:

- 1) ICNN is introduced into the power system transient stability prediction and optimization for the first time. By using ICNN, transient stability constraint can be constructed as a convex constraint to the optimization variables, such as wind farm and synchronous generator active power outputs.
- 2) ICNN-based convex transient stability constraint is combined with the DLPF model to build a novel TSC-OPF and MPC model containing only linear and convex constraints, which can solve the shortcomings of poor convergence and local optimal solution of traditional optimization model and other alternatives.

II. PROBLEM STATEMENT

The purpose of MPC is to establish a control sequence to minimize the whole objective function over a continuous horizon. In this paper, MPC is used to repeatedly solve the multi-period TSC-OPF, which is formulated below [14]–[16].

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A. Objective Function

The objective function is to minimize the total cost of generation and wind curtailment during a control horizon.

$$\begin{aligned} \text{Minimize } & \sum_t^{t+T} \Delta t \left[\sum_{g=1}^{n_g} C_g(P_{g,t}) + \sum_{w=1}^{n_w} C_w(\Delta P_{w,t}) \right] \\ C_g(P_{g,t}) &= a_g P_{g,t}^2 + b_g P_{g,t} + c_{g,t} \\ C_w(\Delta P_{w,t}) &= c_w \Delta P_{w,t} \\ \Delta P_{w,t} &= P_{w,t}^{MPPT} - P_{w,t} \end{aligned} \quad (1)$$

where t denotes current time instant; T denotes the MPC control horizon; Δt denotes the discretized time step; n_g is the number of generators; $P_{g,t}$ denotes the active power output of g -th generator at time t ; $C_g(\cdot)$ is the generation cost function; a_g , b_g , and c_g are cost constants of P_g ; n_w is the number of wind farms; $P_{w,t}$ denotes the active power output of w -th wind farm; $P_{w,t}^{MPPT}$ denotes the maximum active power output of $P_{w,t}$ according to maximum power point tracking (MPPT); $\Delta P_{w,t}$ denotes the curtailed active power of $P_{w,t}$; $C_w(\cdot)$ is the wind curtailment cost function; c_w is cost constant of P_w .

B. Power Balance Constraints

The polar AC power balance constraints are [3]

$$\begin{aligned} P_{i,t} &= \sum_{j=1}^{n_b} G_{ij} V_{i,t} V_{j,t} \cos \theta_{ij,t} + \sum_{j=1}^{n_b} B_{ij} V_{i,t} V_{j,t} \sin \theta_{ij,t} \quad (2) \\ Q_{i,t} &= \sum_{j=1}^{n_b} G_{ij} V_{i,t} V_{j,t} \sin \theta_{ij,t} - \sum_{j=1}^{n_b} B_{ij} V_{i,t} V_{j,t} \cos \theta_{ij,t} \quad (3) \end{aligned}$$

$$i = 1, 2, \dots, n_b$$

where n_b is the number of buses; $P_{i,t}$ and $Q_{i,t}$ are injected active and reactive power at bus i at time t ; $V_{i,t}$ and $V_{j,t}$ are voltage magnitude at bus i and j ; $\theta_{ij,t}$ is bus voltage angle difference between bus i and j ; G_{ij} and B_{ij} are the real and imaginary parts of the bus admittance Y_{ij} ;

C. Transmission Line Power Flow Constraints

The transmission line active power flow inequality constraints are

$$|P_{l,t}| \leq P_l^{\max}, l = 1, 2, \dots, n_l \quad (4)$$

where n_l is the number of transmission lines; $P_{l,t}$ denotes the active power flow at line l ; $P_l^{\max}$ is the max active power flow limits of the lines.

D. Variable Limits

Variable limits include reference bus voltage angle inequality constraint, bus voltage magnitude inequality constraints, generator active and reactive power output constraints, and wind farm active and reactive power output constraints, i.e.,

$$\theta_i^{\text{ref}} \leq \theta_{i,t} \leq \theta_i^{\text{ref}}, i \in \zeta^{\text{ref}} \quad (5)$$

$$V_i^{\min} \leq V_{i,t} \leq V_i^{\max}, i = 1, \dots, n_b \quad (6)$$

$$P_g^{\min} \leq P_{g,t} \leq P_g^{\max}, g = 1, \dots, n_g \quad (7)$$

$$Q_g^{\min} \leq Q_{g,t} \leq Q_g^{\max}, g = 1, \dots, n_g \quad (8)$$

$$P_w^{\min} \leq P_{w,t} \leq P_{w,t}^{MPPT}, w = 1, \dots, n_w \quad (9)$$

$$Q_w^{\min} \leq Q_{w,t} \leq Q_{w,t}^{MPPT}, w = 1, \dots, n_w \quad (10)$$

$$|P_{g,t+1} - P_{g,t}| \leq R_g \Delta t, g = 1, \dots, n_g \quad (11)$$

where ζ^{ref} is the reference bus; $\theta_{i,t}$ and θ_i^{ref} denote the phase angle of reference bus voltage and its rated value; $V_{i,t}$ denotes the i -th bus voltage magnitude; $V_i^{\max}$ and $V_i^{\min}$ are upper and lower limits of V_i ; $P_{g,t}$ and $Q_{g,t}$ denote the g -th generator active and reactive power output; $P_g^{\max}$ and $P_g^{\min}$ are upper and lower limits of P_g ; $Q_g^{\max}$ and $Q_g^{\min}$ are upper and lower limits of Q_g ; $P_{w,t}$ and $Q_{w,t}$ denote the w -th wind farm active and reactive power output; $P_w^{\min}$ and $Q_w^{\min}$ are lower limits of $P_{w,t}$ and $Q_{w,t}$, while $P_{w,t}^{MPPT}$ and $Q_{w,t}^{MPPT}$ are upper limits according to MPPT; R_g denotes the ramp rate of g -th generator [18].

E. Transient Stability Constraint

Transient stability is constrained by DAEs [17].

$$\dot{\mathbf{x}}_t = \mathbf{f}(\mathbf{x}_t, \mathbf{y}_t, \mathbf{u}, \mathbf{p}) \quad (12)$$

$$\mathbf{0} = \mathbf{g}(\mathbf{x}_t, \mathbf{y}_t, \mathbf{u}, \mathbf{p}) \quad (13)$$

where $\mathbf{x}_t$ and $\mathbf{y}_t$ denote state variables and algebraic variables respectively; $\mathbf{u}$ is the input variables; $\mathbf{p}$ denotes the model parameters; $\mathbf{f}(\cdot)$ and $\mathbf{g}(\cdot)$ are nonlinear DAEs. The presence of DAEs makes the solution to the transient stability constraint very time-consuming and subject to scalability issues for large-scale systems. In addition, DAEs-based transient stability constraint is non-linear and non-convex, which may cause the optimization to fall into a local optima or non-convergence.

III. PROPOSED ICNN-BASED TSC-OPF FOR MODEL PREDICTIVE CONTROL

To address the convergence performance issue of the traditional non-linear and non-convex TSC-OPF model and avoid local optima, we integrate the DLPPF and ICNN-based convex transient stability constraint and propose a novel TSC-OPF model. This model only contains convex and linear constraints.

A. Decoupled Linearized Power Flow

The polar AC power balance constraints are (2) and (3). For most scenarios in real power systems, the bus voltage magnitudes are approximately 1 p.u and phase angles are less than 10° . Based on these facts, the DLPPF model is proposed [3]. By using approximation $g_{ij} V_i (V_i - V_j \cos \theta_{ij}) \approx g_{ij} (V_i - V_j)$, (2) and (3) can be approximated to [3]

$$\begin{bmatrix} P \\ Q \end{bmatrix} = - \begin{bmatrix} B' & -G \\ G & B \end{bmatrix} \begin{bmatrix} \theta \\ V \end{bmatrix} \quad (14)$$

where B' is B without shunt elements

$$\theta = [\theta_R^T, \theta_S^T, \theta_L^T]^T \quad (15)$$

$$V = [V_R^T, V_S^T, V_L^T]^T \quad (16)$$

$$Y = \begin{bmatrix} Y_{RR} & Y_{RS} & Y_{RL} \\ Y_{SR} & Y_{SS} & Y_{SL} \\ Y_{LR} & Y_{LS} & Y_{LL} \end{bmatrix} \quad (17)$$

where subscript $\mathcal{R}$, $\mathcal{S}$ and $\mathcal{L}$ denote $V\theta$, PV , and PQ buses, respectively. (14) can be transformed to

$$\begin{bmatrix} \tilde{P} \\ \tilde{Q} \end{bmatrix} = \begin{bmatrix} H & N \\ M & L \end{bmatrix} \begin{bmatrix} \tilde{\theta} \\ \tilde{V} \end{bmatrix} \quad (18)$$

where

$$\begin{bmatrix} \tilde{P} \\ \tilde{Q} \end{bmatrix} = \begin{bmatrix} P_S \\ P_L \\ Q_L \end{bmatrix} + \begin{bmatrix} B'_{SS} & -G_S & -G_{SS} \\ B'_{LR} & -G_{LR} & -G_{LS} \\ G_{LR} & B_{LR} & B_{LS} \end{bmatrix} \begin{bmatrix} \theta_R \\ V_R \\ V_S \end{bmatrix} \quad (19)$$

$$\begin{bmatrix} H & N \\ M & L \end{bmatrix} = - \begin{bmatrix} B'_{SS} & B'_{SL} & -G_{SL} \\ B'_{LS} & B'_{LL} & -G_{LL} \\ G_{LS} & G_{LL} & B_{LL} \end{bmatrix} \quad (20)$$

$$\tilde{\theta} = [\theta_S^T, \theta_L^T]^T \quad (21)$$

$$\tilde{V} = V_L \quad (22)$$

By performing row operations on (18), the final DLPF for voltage and phase angle decoupling are [3]

$$\begin{bmatrix} \tilde{P} - NL^{-1}\tilde{Q} \\ \tilde{Q} - MH^{-1}\tilde{P} \end{bmatrix} = \begin{bmatrix} \tilde{H} & 0 \\ 0 & \tilde{L} \end{bmatrix} \begin{bmatrix} \tilde{\theta} \\ \tilde{V} \end{bmatrix} \quad (23)$$

where

$$\tilde{H} = H - NL^{-1}M \quad (24)$$

$$\tilde{L} = L - MH^{-1}N \quad (25)$$

The bus voltage magnitudes and phase angles can be calculated by

$$\tilde{\theta} = \tilde{H}^{-1}(\tilde{P} - NL^{-1}\tilde{Q}) \quad (26)$$

$$\tilde{V} = \tilde{L}^{-1}(\tilde{Q} - MH^{-1}\tilde{P}) \quad (27)$$

The approximated active power flow of transmission lines are

$$\tilde{P}_l = \frac{g_{ij}}{tap} \left(\frac{V_i}{tap} - V_j \right) - \frac{b_{ij}}{tap} (\theta_i - \theta_j - \theta_s) \quad (28)$$

where ij denotes the from and to bus of line l ; tap denotes the tap ratio of transformers; θ_s is the phase shifting angle; g_{ij} and b_{ij} real and imaginary parts of line admittance y_{ij} .

B. ICNN-based Convex Transient Stability Constraint

In this section, ICNN is proposed to predict the power system's transient stability. This novel method can transform transient stability constraints into convex form and solves quickly. ICNN is developed originally from the traditional fully connected neural network by constraining the network parameters. Compared with other neural networks, the output of ICNN is a convex function of all inputs, which allows for efficient convex optimization for convex inputs [12]. The structure of a k -layers ICNN is shown in Fig. 1, where y is input, and z is the hidden layers vectors for y .

The mathematical expression of ICNN is

$$z_{i+1} = g_i \left(W_i^{(z)} z_i + W_i^{(y)} y + b_i \right) \quad (29)$$

$$f(y; \theta) = z_k \quad (30)$$

where W and b denote the weights and biases of different layers; g is the activation function; f denotes the ICNN model; θ denotes all parameters in the ICNN.

According to [12], the condition that function f is convex to y is that g is convex and non-decreasing and $W^{(z)}$ is non-negative, i.e., the red arrows in Fig. 1 are non-negative.

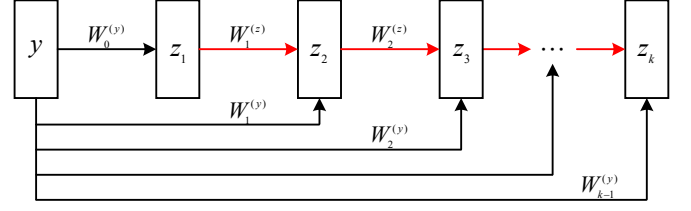


Fig. 1. The model structure of ICNN [12].

addition, we choose Softplus as an activation function, which is convex, non-decreasing, and can derive continuous Jacobian and Hessian matrices. The equation of Softplus is

$$g_i = \log(1 + e^{x_i}) \quad (31)$$

where x_i is the input of Softplus activation function, i.e. $x_i = W_i^{(z)} z_i + W_i^{(y)} y + b_i$. The Jacobian and Hessian matrices of g to x are

$$g'_i = \frac{x'_i}{1 + e^{-x_i}} \quad (32)$$

$$g''_i = \frac{x''_i}{1 + e^{-x_i}} + \frac{x'_i e^{-x_i}}{(1 + e^{-x_i})^2} \quad (33)$$

The Jacobian and Hessian matrix of z_{i+1} to y is

$$z'_{i+1} = g'_i \left(W_i^{(z)} z_i + W_i^{(y)} y + b_i \right) * \left(W_i^{(z)} z'_i + W_i^{(y)} y' \right) \quad (34)$$

$$z''_{i+1} = g''_i \left(W_i^{(z)} z_i + W_i^{(y)} y + b_i \right) * \left(W_i^{(z)} z'_i + W_i^{(y)} y' \right)^2 + g'_i \left(W_i^{(z)} z_i + W_i^{(y)} y + b_i \right) * \left(W_i^{(z)} z''_i \right) \quad (35)$$

Using the above equations, the Jacobian and Hessian matrices of final output to input y can be calculated through chain rule. Jacobian and Hessian matrices are required in some optimization algorithms, such as interior point method, so the proposed model can be solved completely analytically. The Hessian matrix of multivariate convex functions is positive semi-definite matrix, i.e., the eigenvalues are non-negative and the convexity of ICNN can be verified.

Then, we use the proposed ICNN model to predict the power system's transient stability. The input y of ICNN is $[P_w, P_g, P_l, Q_l]$, where P_l and Q_l are active and reactive demands, and the output of ICNN is transient stability index (TSI), which denotes the maximum rotor angle difference of the generators in the transient process [6]

$$TSI = \frac{360^\circ - |\delta_{\max}|}{360^\circ + |\delta_{\max}|} \times 100 \quad (36)$$

$$\delta_{\max} = \max(\delta_{i,t} - \delta_{j,t}) \quad (37)$$

where $\delta_{\max}$ denotes the maximum rotor angle difference between any two generators.

C. DLPF and ICNN-based Model Predictive Control

Based on the DLPF-based linear static stability constraints and ICNN-based convex transient stability constraints, a novel

TSC-OPF model is proposed. The mathematical expression of the proposed method in the form MPC is as follows:

$$\text{Minimize}(1) \quad (38)$$

$$\text{s.t. (7) - (11)} \quad (39)$$

$$\mathbf{V}^{\min} \leq \tilde{\mathbf{V}}_t \leq \mathbf{V}^{\max} \quad (40)$$

$$\left| \tilde{P}_{l,t} \right| \leq P_l^{\max} \quad (41)$$

$$\text{ICNN}(x, y) > \text{TSI}_B \quad (42)$$

where constraints (2), (3) and (5) are always satisfied in DLPF; constraints (7) - (11) remain the same; (27) and (28) are used to construct constraints (40) and (41), respectively. (42) is an ICNN-based transient stability constraint. TSI_B is the boundary between stability and instability. Based on (38) - (42), the proposed TSC-OPF only contains the convex and linear constraints and can find the optimal control strategy by convex optimization.

IV. NUMERICAL RESULTS

The proposed method is tested on the modified IEEE 39-bus system, where four wind farms are located on buses 2, 8, 11, and 21 with a maximum active power output of 225MW. The system model is built in PSS/E. The wind farms use type 3 wind generators, i.e., DFIG. The variation ranges of P_w , P_g , P_l and Q_l are 80% ~ 120% of the rated value. The power factor of loads remains constant. The fault type is set to three-phase short-circuit faults for all credible contingencies. Then, 10000 samples and corresponding TSI are generated by using PSS/E time-domain simulations, 8000 samples of which are training set and 2000 samples being test set.

A. Performance of ICNN-based TSC-OPF model

Based on (29) and (30), an ICNN model is built in PyTorch and is trained with a training set. The optimal number of layers of ICNN is 5; the dimensions of different layers is 52-30-12-5-1; the initial learning rate is 0.06, and the iteration number of training is 300. The test set is used to verify the classification accuracy of ICNN, which is formulated as

$$A_P = \frac{1}{n_t} \sum_{i=1}^{n_t} (\text{TSI}_{P_i} \times \text{TSI}_{T_i} > 0) \quad (43)$$

where A_P is the classification accuracy; n_t denotes the number of test set; TSI_{P_i} and TSI_{T_i} are ICNN-based TSI and time-domain simulations-based TSI, respectively. The sample prediction is correct if TSI_{P_i} and TSI_{T_i} have the same sign (logical equality). The accuracy of the ICNN model is compared against the artificial neural network (ANN) and deep belief network (DBN) model, as shown in Table I, which has the best performance for each method.

TABLE I
COMPARISON OF PERFORMANCE OF DIFFERENT MODELS

Type	Model	A_P	Prediction time
Convex model	ICNN	92.35%	< 0.02s
Non-convex model	ANN	97.45%	
	DBN	98.35%	

It can be observed from Table I that the accuracy of ICNN decreases by 5% ~ 6% compared to DBN and ANN. This is expected because ICNN restricts the range of weights to ensure convexity. To verify this property, the eigenvalues of Hessian matrices of y are calculated, which are all positive. We randomly choose ten scenarios and calculate the eigenvalues of Hessian matrices. The results are shown in Fig. 2.

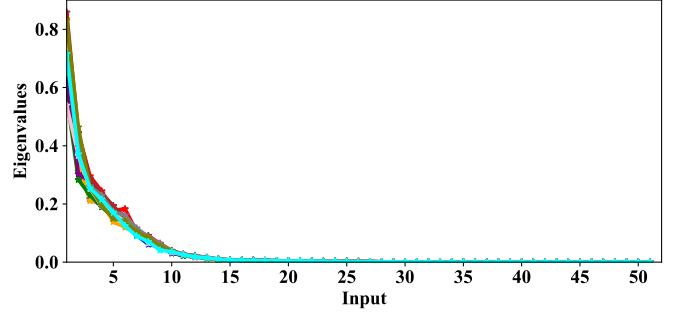


Fig. 2. Eigenvalues of ten random scenarios.

Fig. 2 proves all eigenvalues are non-negative, and thus proves the convexity of ICNN. In addition, the convergence of the proposed method is tested through a large number of unstable scenarios. A comparison with ANN is shown in Table II.

TABLE II
COMPARISON OF CONVERGENCE OF DIFFERENT MODELS

Model	Convergent	Non-convergent	Convergence rate
ICNN	3947	0	100%
ANN	2166	1781	54.8%

From Table II, it can be seen that the proposed ICNN model guarantees 100% convergence, while the traditional non-convex ANN has only 54.8% convergence rate. In order to ensure convergence, some traditional methods do load shedding in presence of divergence, which greatly reduces the reliability of the power supply. The excellent convergence rate of the proposed method implies no load shedding is needed. The training time of ICNN is about 2 min and suitable for the online application.

B. ICNN-based MPC

The ICNN-based MPC is tested with $T = 24h$, $\Delta t = 1h$ and $R_g = 100MW/h$. The wind farm and synchronous generator active power outputs are shown in Fig. 3.

In Fig. 3, the nine purple lines are the active power output of synchronous generators, and the gray line is the slack generator. For four wind farms, the yellow lines and blue lines are active power outputs and maximum active power output according to MPPT, i.e. $P_{w,t}^{MPPT}$, respectively. The black line is the average output of all generators, which has the same shape as the daily load curve. From Fig. 3, we can see that all wind farms and synchronous generators comply with power output constraints and ramp constraints, for example, the power output increment of all generators in

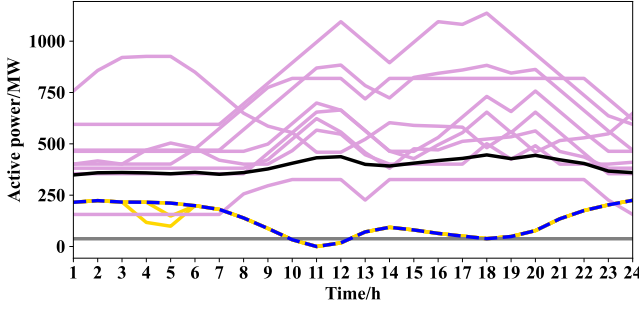


Fig. 3. Wind farm and synchronous generator active power outputs.

adjacent two hours is less than 100MW, and MPC control is achieved. In addition, due to the cost of wind curtailment, the output of wind farms follows the MPPT curve in most hours of the day, and at the 4th and 5th hours, the output is temporarily curtailed to cope with low demands, which shows the feasibility of the proposed method for systems with a high integration of renewable energy. According to the strategy, the generation cost for one day is \$56095.6. The TSI constraints of MPC are plotted in Fig. 4.

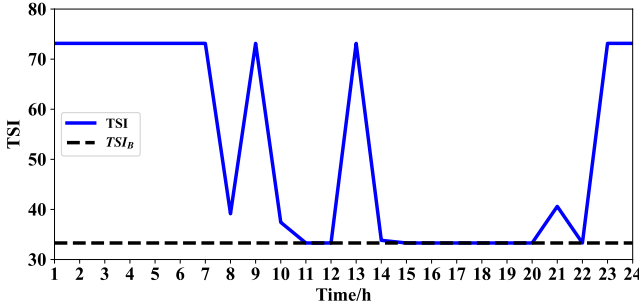


Fig. 4. TSI constraints of MPC.

According to Fig. 4, the $TSI = TSI_B$ at time instants 11, 12, 22 and 14-20 h, which means that the TSC-OPF is constrained by transient stability constraints and ICNN-based constraints work. At other times, MPC is constrained by static constraints and finds the minimum cost. The total optimization time for 24 hours iterations is only 5s, which is very short and good enough in online applications. The proposed method can be performed hourly. Finally, for the sake of comparison, when the ICNN is replaced by ANN in the same model, the probability of simultaneous convergence of ANN over 24 hours is almost 0, due to the poor convergence rate of ANN (i.e., 54.8%). Therefore, it's difficult to apply ANN to MPC.

C. Comparison with AC-OPF Methods

This section compares the proposed method to the AC-OPF method, which determines the transient stability constraint based on time-domain simulations. Because the AC-OPF is difficult to converge for a 24h MPC, 24 1h MPC is performed for 1 day. For AC-OPF, only 1 ~ 4, 6 ~ 8, 13, 14, 23 and 24h can converge, with an average cost of \$2346.5. The average

cost for the proposed method at the same time is only \$2098.5, demonstrating the advantages of the proposed method.

V. CONCLUSION

This paper proposed an analytic ICNN-based MPC method for transient stability enhancement. ICNN is used to substitute the non-linear transient stability constraint, and a novel TSC-OPF model is proposed, which only contains linear and convex constraints with 100% convergence rate. Finally, the applicability of the proposed method to MPC shows that hour-by-hour optimal control within a day can be achieved.

REFERENCES

- [1] R. Madani, S. Sojoudi and J. Lavaei, "Convex relaxation for optimal power flow problem: mesh networks," *IEEE Trans. Power Syst.*, vol. 30, no. 1, pp. 199-211, Jan. 2015.
- [2] C. Coffrin, H. L. Hijazi and P. Van Hentenryck, "The QC relaxation: A theoretical and computational study on optimal power flow," *IEEE Trans. Power Syst.*, vol. 31, no. 4, pp. 3008-3018, July 2016.
- [3] J. Yang, N. Zhang, C. Kang and Q. Xia, "A state-independent linear power flow model with accurate estimation of voltage magnitude," *IEEE Trans. Power Syst.*, vol. 32, no. 5, pp. 3607-3617, Sept. 2017.
- [4] B. Stott, J. Jardim and O. Alsac, "DC power flow revisited," *IEEE Trans. Power Syst.*, vol. 24, no. 3, pp. 1290-1300, Aug. 2009.
- [5] S. M. Fatemi, S. Abedi, G. B. Ghahrepetian, S. H. Hosseini and M. Abedi, "Introducing a novel DC power flow method with reactive power considerations," *IEEE Trans. Power Syst.*, vol. 30, no. 6, pp. 3012-3023, Nov. 2015.
- [6] T. Su, Y. Liu, J. Zhao, and J. Liu, "Deep belief network enabled surrogate modeling for fast preventive control of power system transient stability," *IEEE Trans. Ind. Inform.*, vol. 18, no. 1, pp. 315-326, Jan. 2022.
- [7] Q. Wang and V. Ajjarapu, "PA novel approach to implement generic load restoration in continuation-based quasi-steady-state analysis," *IEEE Trans. Power Syst.*, vol. 20, no. 1, pp. 516-518, Feb. 2005.
- [8] P. Bhui and N. Senroy, "Real time prediction and control of transient stability using transient energy function," *IEEE Trans. Power Syst.*, vol. 32, no. 2, pp. 923-934, March 2017.
- [9] H. Yuan and Y. Xu, "Preventive-corrective coordinated transient stability dispatch of power systems with uncertain wind power," *IEEE Trans. Power Syst.*, vol. 35, no. 5, pp. 3616-3626, Sep. 2020.
- [10] S. A. Siddiqui, K. Verma, K. R. Niazi, and M. Fozdar, "Real-time monitoring of post-fault scenario for determining generator coherency and transient stability through ANN," *IEEE Trans. Ind. Appl.*, vol. 54, no. 1, pp. 685-692, Jan. 2018.
- [11] T. Su, Y. Liu, J. Zhao, and J. Liu, "Probabilistic stacked denoising autoencoder for power system transient stability prediction with wind farms," *IEEE Trans. Power Syst.*, vol. 36, no. 4, pp. 3786-3789, Jul. 2021.
- [12] B. Amos, L. Xu, and J. Z. Kolter, "Input convex neural networks," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2017, pp. 146-155.
- [13] Y. Chen, Y. Shi, and B. Zhang, "Optimal control via neural networks: A convex approach," in *Proc. Int. Conf. Learning Representation, (ICLR)*, 2019, pp. 1-13.
- [14] J. A. Martin and I. A. Hiskens, "Corrective model-predictive control in large electric power systems," *IEEE Trans. Power Syst.*, vol. 32, no. 2, pp. 1651-1662, March 2017.
- [15] G. Qiu *et al.*, "Analytic deep learning-based surrogate model for operational planning with dynamic TTC constraints," *IEEE Trans. Power Syst.*, vol. 36, no. 4, pp. 3507-3519, Jul. 2021.
- [16] A. Di Giorgio, F. Liberati, A. Lanna, A. Pietrabissa and F. D. Priscoli, "Model predictive control of energy storage systems for power tracking and shaving in distribution grid," *IEEE Trans. Sustain. Energy*, vol. 8, no. 2, pp. 496-504, April 2017.
- [17] J. Zhao, *et al.*, "Power system dynamic state estimation: motivations, definitions, methodologies, and future work," *IEEE Trans. Power Syst.*, vol. 34, no. 4, pp. 3188-3198, Jul. 2019.
- [18] E. Moiseeva, M. R. Hesamzadeh and D. R. Biggar, "Exercise of market power on ramp rate in wind-integrated power systems," *IEEE Trans. Power Syst.*, vol. 30, no. 3, pp. 1614-1623, May 2015.